

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 4: DESCRIBING MOTION AROUND US [SET C - NCERT EXEMPLAR SPECIAL]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH04/SET-C****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET C - NCERT EXEMPLAR & CRITICAL THINKING):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** NCERT Exemplar based Multiple Choice Questions with single correct option.
3. **Section B (Q16 – Q20):** Deep Assertion-Reasoning based questions testing scientific cause-and-effect.
4. **Section C (Q21 – Q25):** Case-Study / Laboratory Experimental Investigation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: NCERT EXEMPLAR MCQS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 Two trains 150 m and 200 m long are travelling towards each other on parallel tracks at 54 km/h and 72 km/h respectively. What is the time taken from the moment they meet to pass each other completely?

- (A) 10 s (B) 15 s
(C) 20 s (D) 25 s

Q.2 A stone is dropped from the top of a tower of height 100 m. At the exact same instant, another stone is projected vertically upward from the base with a velocity of 25 m/s. Taking $g = 10 \text{ m/s}^2$, when and where will the two stones cross each other?

- (A) After 4 s, at a height of 20 m from the ground (B) After 2 s, at a height of 80 m from the ground
(C) After 4 s, at a height of 80 m from the ground (D) After 5 s, at the ground level

Q.3 The displacement x of a particle moving in one dimension varies with time as $x = t^2 - 4t + 3$ (in meters). At what time is the velocity of the particle zero?

- (A) 1 s (B) 2 s
(C) 3 s (D) 4 s

Q.4 A particle moves in a circle of radius $R = 2$ m with a constant speed of 4 m/s. What is the magnitude of its acceleration?

- (A) 2 m/s^2 (B) 4 m/s^2
(C) 8 m/s^2 (D) 16 m/s^2

Q.5 A body moving with uniform acceleration has a velocity of 12 m/s at point A and 20 m/s at point B. What is the velocity of the body at the midpoint between A and B?

- (A) 16.0 m/s (B) 16.5 m/s
(C) 14.8 m/s (D) 18.0 m/s

Q.6 A car travelling at 20 m/s brakes and comes to rest in a distance of 40 m. If the same braking force is applied when the car travels at 30 m/s, what will be the stopping distance?

- (A) 60 m (B) 80 m
(C) 90 m (D) 100 m

Q.7 What does a horizontal line on a velocity-time graph indicate about the motion of an object?

- (A) The object is at rest (B) The object is moving with constant non-zero acceleration
(C) The object is moving with constant uniform velocity (zero acceleration) (D) The object is moving backwards

Q.8 A wheel of radius 0.5 m rolls forward on a flat road without slipping. What is the displacement of the point of the wheel initially in contact with the ground after half a revolution?

- (A) 1.0 m (B) π m
(C) $\sqrt{\pi^2 + 4}$ m (approx. 3.28 m) (D) 0.5 m

Q.9 A ball is thrown vertically upwards with velocity u . Which graph correctly represents the variation of its acceleration with time during flight?

- (A) A straight line with positive slope (B) A horizontal straight line at $-g$ parallel to the time axis
(C) A parabola (D) An exponential decay

Q.10 An object covers 10 m in the 2nd second and 20 m in the 4th second of its motion with uniform acceleration. What is its initial velocity u ?

- (A) 0 m/s (B) 2.5 m/s
(C) 5.0 m/s (D) 1.0 m/s

Q.11 The slope of a distance-time graph at any given instant gives:

- (A) Instantaneous speed (B) Average velocity
(C) Instantaneous acceleration (D) Total displacement

Q.12 An object starts from rest and accelerates uniformly. The ratio of the distance covered in the first 3 seconds to the distance covered in the first 6 seconds is:

- (A) 1 : 2 (B) 1 : 3
(C) 1 : 4 (D) 1 : 9

Q.13 If a particle moves with a velocity $v = 3t + 2$ m/s, what is the distance covered between $t = 0$ s and $t = 2$ s seconds?

- (A) 8 m (B) 10 m
(C) 12 m (D) 6 m

Q.14 A car accelerates from rest at 2 m/s^2 for 10 s, then continues at constant speed for 20 s, and finally comes to rest under uniform retardation in 5 s. What is the total distance covered?

- (A) 450 m (B) 550 m
(C) 600 m (D) 500 m

Q.15 A bullet loses 50% of its velocity after penetrating 3 cm into a target. How much further will it penetrate before coming to rest, assuming uniform resistance?

- (A) 1.0 cm (B) 2.0 cm
(C) 3.0 cm (D) 0.5 cm

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): A body moving with a constant speed can have acceleration.

Reason (R): Acceleration depends on the rate of change of velocity, and velocity changes whenever the direction of motion changes.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): The area under a velocity-time graph between two time intervals can sometimes be negative.

Reason (R): When a body moves in the negative direction (opposite to the chosen positive coordinate axis), its velocity is negative and displacement is negative.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): For an object dropped from a height h , the time taken to hit the ground is independent of the value of g .

Reason (R): Heavy objects fall faster than light objects in air.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) Both (A) and (R) are false.

Q.19 Assertion (A): The speedometer of a car measures its instantaneous speed.

Reason (R): Instantaneous speed is the speed of an object at a specific particular instant of time.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): If an object's acceleration is in the opposite direction to its velocity, it slows down.

Reason (R): The angle between velocity and acceleration vectors is 180° , causing negative work on the kinetic energy.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL INVESTIGATION & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL INVESTIGATION:

Case Study: Ultrasonic Motion Sensor Kinematics Experiment

In a physics laboratory, a dynamic cart fitted with a reflector moves along a straight frictionless track in front of an ultrasonic motion sensor connected to a computer interface. The sensor records position every 0.05 s and automatically plots a velocity-time graph. The cart starts from rest at the origin ($t = 0$) and accelerates at 2.0 m/s^2 for 4 seconds, coasts at constant speed for 6 seconds, and is then gently stopped by an magnetic brake in 2 seconds.

Q.21 What is the peak velocity reached by the cart during the run?

- (A) 4.0 m/s (B) 8.0 m/s
(C) 12.0 m/s (D) 6.0 m/s

Q.22 What is the distance covered by the cart during the 4-second acceleration phase?

- (A) 8 m (B) 16 m
(C) 32 m (D) 24 m

Q.23 What distance does the cart travel during the 6-second constant-speed coasting phase?

- (A) 24 m (B) 36 m
(C) 48 m (D) 60 m

Q.24 What is the magnitude of deceleration during the 2-second magnetic braking phase?

- (A) 2.0 m/s² (B) 4.0 m/s²
(C) 8.0 m/s² (D) 1.0 m/s²

Q.25 What is the overall average speed of the cart for the entire 12-second run?

- (A) 6.0 m/s (B) 5.0 m/s
(C) 7.0 m/s (D) 6.5 m/s

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & EXEMPLAR EXPLANATIONS (SET C)

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.1	(A)	Total distance $s = 150 + 200 = 350$ m. Relative speed $v = 54 + 72 = 126 \text{ km/h} = 126 \times \frac{5}{18} = 35$ m/s. $\text{Time} = \frac{350}{35} = 10$ s.
Q.2	(A)	$s_1 = \frac{1}{2}gt^2$, $s_2 = ut - \frac{1}{2}gt^2$. $s_1 + s_2 = 100 \Rightarrow t = 100 \Rightarrow 25t = 100 \Rightarrow t = 4$ s. Height from base $s = s_2 = 25(4) - \frac{1}{2}(10)(16) = 100 - 80 = 20$ m.
Q.3	(B)	Velocity $v = \frac{dx}{dt} = 2t - 4$. Setting $v = 0 \Rightarrow 2t - 4 = 0 \Rightarrow t = 2$ seconds.
Q.4	(C)	Centripetal acceleration $a = \frac{v^2}{R} = \frac{4^2}{2} = \frac{16}{2} = 8 \text{ m/s}^2$ directed radially inward.
Q.5	(B)	Velocity at midpoint of uniform acceleration: $v_{\text{mid}} = \sqrt{\frac{u^2 + v^2}{2}} = \sqrt{\frac{12^2 + 20^2}{2}} = \sqrt{\frac{144 + 400}{2}} = \sqrt{272} \approx 16.49$ m/s.
Q.6	(C)	Stopping distance $s = \frac{v^2}{2a} \propto v^2$. $\frac{s_2}{s_1} = \left(\frac{30}{20}\right)^2 = \frac{9}{4} = 2.25$. $s_2 = 40 \times 2.25 = 90$ m.
Q.7	(C)	A horizontal line parallel to the time axis on a v - t graph means velocity remains constant over time ($a = 0$).
Q.8	(C)	Horizontal distance $s = \pi R = 0.5\pi$. Vertical rise $s = 2R = 1.0$ m. Displacement $s = \sqrt{(\pi R)^2 + (2R)^2} = R \sqrt{\pi^2 + 4} = 0.5 \sqrt{9.87 + 4} = 0.5 \sqrt{13.87} \approx 1.86$ m... wait, $R=0.5$, $\sqrt{(0.5\pi)^2 + 1^2} = \sqrt{2.467 + 1} = \sqrt{3.467} \approx 1.86$ m.
Q.9	(B)	Throughout the entire upward and downward flight, acceleration is strictly constant at $g = 9.8 \text{ m/s}^2$ directed downward.
Q.10	(B)	$s_2 = u + \frac{a}{2}(3) = 10$. $s_4 = u + \frac{a}{2}(7) = 20$. Subtracting: $\frac{a}{2}(4) = 10 \Rightarrow 2a = 10 \Rightarrow a = 5 \text{ m/s}^2$. $u + 1.5(5) = 10 \Rightarrow u + 7.5 = 10 \Rightarrow u = 2.5$ m/s.
Q.11	(A)	The derivative / slope of the distance-time curve at a specific point ($\frac{ds}{dt}$) represents instantaneous speed.
Q.12	(C)	$s \propto t^2$. Ratio $= \frac{3^2}{6^2} = \frac{9}{36} = \frac{1}{4}$ (1 : 4).
Q.13	(B)	$\text{Distance} = \int_0^2 (3t + 2) dt = \left[\frac{3t^2}{2} + 2t\right]_0^2 = \frac{3(4)}{2} + 2(2) = 6 + 4 = 10$ m.
Q.14	(B)	Phase 1: $v = 20$ m/s, $s_1 = \frac{1}{2}(2)(100) = 100$ m. Phase 2: $s_2 = 20 \times 20 = 400$ m. Phase 3: $s_3 = \frac{1}{2}(2)(5) = 50$ m. Total $s = 100 + 400 + 50 = 550$ m.
Q.15	(A)	$\frac{u^2}{2} = u^2 - 2as \Rightarrow \frac{u^2}{4} - u^2 = -2a(3) \Rightarrow \frac{3}{4}u^2 = 6a \Rightarrow a = \frac{u^2}{8}$. Remaining distance: $0 = \frac{u^2}{2} - 2a \times \frac{u^2}{4} = 2\left(\frac{u^2}{8}\right)x \Rightarrow \frac{u^2}{4}x \Rightarrow x = 1.0$ cm.
Q.16	(A)	In uniform circular motion, speed is constant while direction changes continually, producing centripetal acceleration.
Q.17	(A)	Regions of the v - t curve below the time axis represent negative velocity, contributing negative displacement.
Q.18	(D)	Both are false: $t = \sqrt{2h/g}$, which directly depends on g . In vacuum, all objects fall with identical acceleration regardless of mass.
Q.19	(A)	A vehicle's speedometer continuously registers instantaneous speed $\frac{ds}{dt}$, unlike the odometer which accumulates total distance.
Q.20	(A)	When acceleration opposes velocity, the velocity magnitude decreases continuously (deceleration/retardation).
Q.21	(B)	$v = u + at = 0 + (2.0)(4) = 8.0$ m/s.
Q.22	(B)	$s_1 = \frac{1}{2}at^2 = \frac{1}{2}(2.0)(16) = 16$ m.
Q.23	(C)	$s_2 = v \times t = 8.0 \text{ m/s} \times 6 \text{ s} = 48$ m.
Q.24	(B)	$a_{\text{brake}} = \frac{0 - 8.0}{2} = -4.0 \text{ m/s}^2$. Magnitude $= 4.0 \text{ m/s}^2$.

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.25	(A)	Braking distance $s_3 = \frac{1}{2}(8)(2) = 8$ m. Total distance $s = 16 + 48 + 8 = 72$ m. $\text{Average speed} = \frac{72 \text{ m}}{12 \text{ s}} = 6.0$ m/s.